Digital Operations Analytics

PORTFOLIO CASE STUDY | NOVEMBER 2026

Objective: Identify which fictional business services may need follow-up by comparing incident records, daily error rates and service-level thresholds. The data and findings are synthetic.

Project at a glance

8	120	15	5/5
Applications	Daily records	Incident records	Checks passing

Approach

• Loaded and cleaned CSV records with Pandas, then calculated request-weighted error rates and service KPIs.

• Used SQL filters, grouping and joins in SQLite to compare incident activity across applications.

• Created a Streamlit dashboard with daily service trends, incident filters and CSV export.

Key findings

Payroll & HR Portal: Three P2 incident records with 12,801 summed affected-user estimates (not necessarily unique users). Its daily error rate on 3 November was 2.45%, compared with a synthetic threshold of 1.00%.

Engineering PLM: Daily error rate of 1.74% on 3 November, compared with a synthetic threshold of 1.50%. Overall request-weighted error rate across the simulated dataset was 0.275%.

Interpretation and recommendation

Payroll and Engineering are candidates for further investigation based on the configured thresholds. Before diagnosing a real incident, an analyst should check consistent observation windows, logs, system dependencies and unique user impact. The daily telemetry and incident examples were produced separately, so no causal relationship can be confirmed.

Scope and limitations

This is a sample-data analysis dashboard, not a live monitoring or incident response system. Its purpose is to demonstrate data preparation, SQL investigation, service reporting and evidence-based interpretation.

Technology: Python | Pandas | SQLite / SQL | Streamlit | Plotly